

Metals and Non-metals (Set B) — MCQ Worksheet

Science • Chapter 3 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. Which property is not typical of metals?

- (A) Lustre
- (B) Malleability
- (C) Ductility
- (D) Brittleness

Q2. Which non-metal is essential for life?

- (A) Sulphur
- (B) Phosphorus
- (C) Oxygen
- (D) Argon

Q3. Aluminium foil is malleable, indicating:

- (A) Can be drawn into wires
- (B) Can be hammered into sheets
- (C) Conducts electricity
- (D) Is heavy

Q4. An alloy of iron and carbon (low percentage) is:

- (A) Steel
- (B) Brass
- (C) Bronze
- (D) Solder

Q5. Aluminium reacts with oxygen to form a protective layer of:

- (A) Al_2O_3
- (B) AlO
- (C) $\text{Al}(\text{OH})_3$
- (D) AlCl_3

Q6. Most reactive metal in activity series is:

- (A) Potassium
- (B) Iron
- (C) Copper
- (D) Silver

Q7. Reaction of metal with oxygen gives:

- (A) Metal oxide (mostly basic)
- (B) Acid
- (C) Metal carbonate
- (D) Metal hydroxide

Q8. Reaction of metal with water releases:

- (A) Hydrogen gas
- (B) Oxygen
- (C) Chlorine
- (D) Carbon dioxide

Q9. An alloy of copper and tin is:

- (A) Bronze
- (B) Brass
- (C) Steel
- (D) Solder

Q10. Carbon as graphite is used in:

- (A) Pencils, electrodes
- (B) Cooking
- (C) Construction
- (D) Cosmetics

Q11. Reactivity series puts least reactive metal at:

- (A) Top
- (B) Bottom
- (C) Middle
- (D) Random

Q12. Gold ornaments are mostly alloy because:

- (A) Pure gold is too soft
- (B) Pure gold reacts with air
- (C) Cost reduction
- (D) Both A and C

Q13. Stainless steel contains:

- (A) Iron, chromium, nickel and carbon
- (B) Copper, zinc
- (C) Lead, tin
- (D) Aluminium, magnesium

Q14. An electrolytic process used to coat one metal on another is:

- (A) Electroplating
- (B) Roasting
- (C) Calcination
- (D) Smelting

Q15. Aluminium displaces ____ from its oxide on heating:

- (A) Iron from iron oxide
(C) Calcium

- (B) Sodium
(D) Potassium

Q16. Why is sodium kept in kerosene?

- (A) It reacts vigorously with air and water
(C) Looks better

- (B) It is too valuable
(D) It absorbs water

Q17. Roasting is heating a sulphide ore in:

- (A) Excess air
(C) Hydrogen

- (B) Absence of air
(D) Sulphur

Q18. Calcination is heating a carbonate ore in:

- (A) Absence of air
(C) Hydrogen

- (B) Excess oxygen
(D) Steam

Q19. Sodium and potassium are extracted by:

- (A) Electrolysis of molten chlorides
(C) Roasting

- (B) Reduction with carbon
(D) Calcination

Q20. Why does copper not react with dilute HCl but reacts with concentrated HNO_3 ?

- (A) Copper is below hydrogen in activity series; HNO_3 is an oxidising acid
(C) Concentrated HCl is unavailable

- (B) HCl is too weak
(D) HNO_3 is base

Q21. The thermite reaction ($\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow 2\text{Fe} + \text{Al}_2\text{O}_3$) is used to:

- (A) Weld broken railway tracks (produces molten iron)
(C) Generate electricity

- (B) Cook food
(D) Bleach clothes

Q22. Two metals X and Y are placed in CuSO_4 solution separately. X displaces copper, Y does not. The order of reactivity is:

- (A) X is more reactive than Cu and Y
(C) $X = Y$ in reactivity

- (B) Y is more reactive than X
(D) Cannot determine

Q23. Why does silver chloride turn dark in sunlight?

- (A) It decomposes to silver and chlorine
(C) Water absorbs it

- (B) Air oxidises it
(D) It is hygroscopic

Q24. Anodising is done on aluminium to:

- (A) Increase the thickness of oxide layer for protection
(C) Make it conductive

- (B) Increase weight
(D) Polish it

Q25. When sodium is added to water, the gas evolved catches fire because:

- (A) Reaction is highly exothermic and hydrogen ignites
(C) Water heats spontaneously

- (B) Sodium produces oxygen
(D) Sodium absorbs heat